

FORENSIC

SCIENCE HYPOTHETICAL

THE BURGLARY IN SUTTON STREET

JEANNETTE JOLLEY



Quality ready-to-use
resources



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Other titles by Jeannette Jolley

- Forensic Science Hypothetical
 - The Kidnapping of Jamie Lawson
 - The Death of Barnaby Hill
- Forensic Science for High Schools (Books 1–4)
- Everyday Science (Books 1–2)

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Introduction

The Burglary in Sutton Street is an extended practical project in which students are the detectives who must solve the crime. Solving this crime does not require any previous study or formal knowledge of forensic science.

This book is aimed at junior secondary level, years 7–8; however, it is also suitable for upper primary students in need of a challenge. This project and the exercises can be used to teach a topic on forensic science, or as an engaging exercise to fill in a gap of several weeks when core work has been completed.

Teaching instructions

Six separate exercises are provided at the beginning of the book. You can use these exercises to introduce the topic of forensic science in general, or to prepare students for the individual skills they will use during the extended project.

It is strongly recommended that students work in groups to solve the crime. Each group should record their work and conclusions in the Investigative Report, provided on pages 36–45. Three students has been found to be the optimal group size for ensuring that each student stays involved and can make a useful contribution. You could have the students in a group take turns completing each part of the report. Some students like to develop a competition between the groups, aiming to be the first to complete the work and solve the crime correctly.

It is estimated that solving the crime will take about six 50-minute lessons for a class of average 12 year olds. This allows time for discussing relevant issues such as personal safety and the risk of contaminating evidence if students should ever be first to arrive at a crime scene. To extend the topic further, you could show some appropriate videos or facilitate class discussion about the use of forensic science in popular television shows.

Summary of each activity and preparation required

Exercise 1 – Searching the crime scene (page 6) This activity helps the students understand how a crime scene is searched, and how evidence is collected by the forensic crime scene examiners.

Preparation: Photocopy sheet.

Exercise 2 – Fingerprints (page 7) This activity introduces the concept of fingerprints.

Preparation: Photocopy sheet.

Exercise 3 – Elimination fingerprint form (page 8) Students use inked pads to make their own fingerprints. It is suggested they practise first on scrap paper before completing the form. Too much ink on the fingers or pressing too hard will result in blurred images.

Preparation: Provide scrap paper, a photocopy of the form, a magnifying lens and an ink pad (available from stationary suppliers or newsagencies). Old newspapers, soap and water are also recommended.

Exercise 4 – Dusting fingerprints (page 9) In this activity, students will collect fingerprints, experimenting with different surfaces and powders. It is lots of fun for the students, but can be quite messy to clean up. Students will then consider how fingerprints help in solving crimes.

Preparation: Black and white powders are needed for dusting. For the white powder, use talcum powder. For the black powder, use fine carbon powder. Fingerprints can be made on any hard, shiny surface. For a white surface, a white board is ideal. Ceramic surfaces will also give clear prints. Small, glossy, black and white bathroom tiles can be purchased from a hardware store. To make a cheap fingerprint 'brush', wrap a ball of cotton wool in a scrap of gauze and tie it onto a wooden icecream stick. It is essential that brushes used for black powder are kept separate from those used for white powder. Squares of black and white paper and sticky tape are also required for each group.

Exercise 5 – Matching fingerprints (page 10) Students learn about the small characteristics used to match fingerprints. They inspect the fingerprints provided, then their own.

Preparation: Provide magnifying lenses.

Exercise 6 – Making your own lip prints (page 11) This is a practical activity where students use lipstick to create their own lip prints to examine.

Preparation: Because of the risk of infection, all students should bring their own lipsticks from home. Dark colours work best. The teacher should provide a sheet of clean white paper for each student, and a supply of cleansing cream to remove the lipstick. Magnifying lenses are also recommended.

Summary of hypothetical crime and preparation required

Initial preparation: Photocopy the Investigative Report (pages 36–45) and provide as a booklet for each group. An extra blank or lined sheet can be added inside the back cover for students to record any additional information. Photocopy the information sheets (pages 12–35), with each part on different coloured sheets. Note that Parts 5 and 6 are on the same page, and that Part 8, the fingerprints, are best copied onto white paper. These sheets should be given out to the students in numeric order as the previous part is completed. As students work through each part, earlier parts should be available so that they can recheck information.

Part 1 – A crime is discovered (page 12)

Two children arrive home from school to discover their home has been burgled.

Preparation: The teacher should introduce the activity with a class discussion about what students should do if they come home to a burglary, like the twins did. Further discussion could include what to do if they either see a crime being committed, or are the first to discover a crime scene.

Part 2 – Evidence found at 15 Sutton Street (pages 13–14)

These pages describe the initial evidence found at the crime scene. Students use the information to answer the questions in their report booklet. No preparation is required.

Part 3 – Lip prints (page 15)

Students are provided with more information about lip prints and introduced to the concept of comparing lip prints to solve a crime. No preparation is required.

Part 4 – Analysing paint chips and glass (page 16)

Paint chips: The students use a magnifying lens to examine the layers in six different paint flakes to see which one matches the seventh flake, which was collected from the crime scene.

Preparation: There are two possible ways to acquire six different blue paint flakes. The simplest is to visit a large panelbeating business and ask them to collect blue paint flakes from six different cars over a period of time. Alternatively, paint flakes can be made in the science laboratory. You will need some small tins of oil based or acrylic paint (including at least one blue) and a sheet of plastic, such as a rubbish bin liner. Preparation of the paint flakes will take several days, allowing the layers to dry overnight, so sufficient time must be allowed to prepare these before they are required. However, once done, they can be reused over several years.

- Paint six patches on the plastic using blue paint (one or different blue colours can be used).
- When dry, paint a second coat over the first using different colours. Try to vary the colours.
- Add a third layer in a different colour, making sure that no two patches have exactly the same combination of colours. A fourth layer can be added to some or all patches.
- When the patches have dried, break them up into pieces and store them in labelled envelopes.
- Use a small amount of plasticine to mount six different flakes on a microscope slide. One edge of each flake should be embedded in plasticine, with another edge pointing upwards so the students can inspect the layers. Label the flakes on the slide A, B, C, D, E and F.
- On a separate microscope slide, mount a flake that matches sample C and label it 'Found at crime scene'.

Glass: The students compare two pieces of glass found at the crime scene with six broken headlights, to determine if the two pieces have come from a suspect's car. They are asked to consider if matching the glass actually proves that the car's owner was driving the car at the time the glass was lost at the crime scene.

Preparation: Provide small squares of tracing paper to trace the two pieces of glass.

Part 5 – Analysing fibres (page 17)

The students use a magnifying lens to inspect a range of different red fibres such as cotton, silk, polyester, nylon, wool and mohair. Some students may need help with question 2 in the Investigative Report (p.40). Question 3 will require some research in an encyclopedia.

Preparation: The teacher should collect five different types of dark red fibres, including wool, and place them in five separate clear bags labelled with the type of fibre. Alternatively, the samples could be

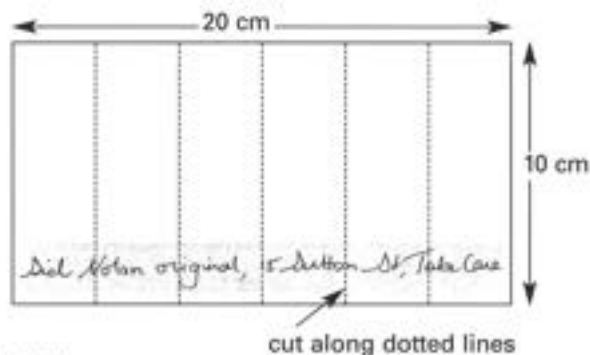
mounted on labelled microscope slides using clear nail polish. Some of the wool should be placed into a sixth bag and labelled 'Unknown fibre found at the crime scene'. Provide a microscope or magnifying lens.

Part 6 – Crumpled note and chromatography (page 17)

This is a practical activity involving paper chromatography. Students use absorbent paper to compare the colour patterns (chromatograms) from six different pens with the colour pattern from part of the note. Younger students may need some help in cutting out the oval sections as shown in the diagram on page 17. A demonstration of how this can be done would be helpful.

Preparation:

- Purchase or find six different black pens that give different coloured patterns when tested with water. Water-soluble overhead markers and fine, felt-tipped pens give the most interesting arrays of colours. Label each pen with one of the six suspect's names.
- Obtain some sheets of blotting or chromatography paper. Each group will need one piece measuring about 15 cm × 10 cm.
- Take another piece of blotting paper, 20 cm × 10 cm. Using the pen labelled 'Daniel Davis', copy the handwriting on the crumpled note about 1 cm from the bottom, as shown at right. (For authenticity in handwriting style, you could trace the note supplied in the evidence.) This paper is cut into vertical strips in front of the class so that each group gets part of the note to test (at least 3 or 4 letters each).
- Provide two notes if you have a larger class.
- Provide a small beaker for each group in which to test their ink.



Part 7 – Handwriting (page 18)

The students inspect the six different samples of handwriting and decide which one most closely resembles that on the note. They then record the similarities and consider if this proves that the owner of the pen wrote the note. Some class discussion about forgeries could be appropriate here. No other preparation is required.

Part 8 – Fingerprints (pages 19–32)

Students must now compare the fingerprints found at the crime scene (shown in Part 2), with the rolled fingerprints collected from the suspects by the police.

Preparation: The 14 pages of rolled elimination fingerprints should be photocopied together on white paper. A magnifying lens for each group should also be provided.

Part 9 – Shoe prints (page 33)

This page involves comparing a photograph of a shoe print with five prints made from shoes owned by the suspects. Again, the students are asked if this proves the guilt of the shoe's owner, or only strengthens the case against this person. There is no teacher preparation needed for this activity.

Part 10 – Using identikit (page 34)

The students need to read the description of the suspects then select the face that most closely resembles each description. No preparation is required.

Part 11 – Questioning the suspects (page 35)

The students read the final information provided by the suspects and finalise who contributed what evidence. No further preparation is required.

Part 12 – Summary of all the evidence (report pages 44–45)

The students summarise all they have discovered about the crime, including the how and the why. No other preparation is required.

Forensic Science crossword (page 46)

This is a review activity covering aspects of forensic science that students have used and explored, as well as basic details about the burglary in Sutton Street.

Searching the crime scene .. " .. " .. " .. " .. " .. " .. EXERCISE 1

Name _____

When a crime has been committed, there are specially trained police officers who are called out to search a crime scene for clues. These officers are called crime scene examiners (CSEs). They look for evidence (evidence is what the clues are called) that untrained people might not notice. These officers wear gloves, and overalls made of a fabric that does not shed its fibres. Often they also wear elastic caps and overshoes, similar to those worn in hospital operating theatres, so that they don't leave any small particles (trace evidence) from themselves at the scene.

The CSE will usually photograph the scene first, then use a strong torch to inspect the scene in more detail. Tweezers are used to pick up small items that may have been left behind by the criminal. If a smudge of fingerprints is found on a hard surface, it will be dusted using a special powder.

Any shoe prints or impressions left by tyres at the scene are photographed, then a plaster cast is made of each impression. These photographs and casts can then be compared to prints made from the suspect's shoes or tyres.

Sometimes hairs or tiny fibres from a criminal's clothes are found. These are collected and sealed in specially labelled bags, and then sent to the forensic laboratory for further testing.



The police hope that this evidence will lead them to the person or people who carried out the crime, and will provide enough information to convict the guilty person(s) 'beyond all reasonable doubt'.

- 1 Explain why a CSE wears overalls, hair nets and gloves when at a crime scene.

- 2 Why are photographs taken before any evidence is moved?

- 3 Why do you think the CSE uses a torch to search for clues?

- 4 What is done to preserve the information when shoe prints or tyre prints are found?

- 5 What happens to any small items that are found, such as hairs and fibres?

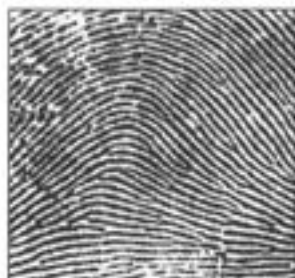
EXERCISE 2

Name

Fingerprints are the smudges left behind by the ridges on our hands. The marks are made by a mixture of oil and sweat that is produced by our skin. When we touch an object, we leave some of this liquid behind in the same pattern as the ridges on our hands. These ridges are unique to every individual in the world.

If they are on a smooth surface, the most common way to make these patterns of ridges visible is to dust them with powder. The powder sticks to the marks. Then sticky tape is placed over the powdered marks and they can be lifted and stuck onto a prepared card of a contrasting colour.

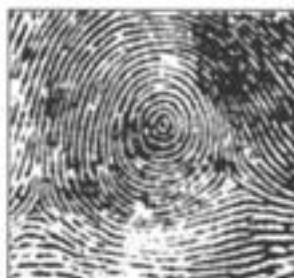
There are four main fingerprint patterns:



arch



loop



whorl



composite
(mixture of patterns)

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

- 1 Above is a set of ten fingerprints. Under each fingerprint, write the type of pattern you think it is.
- 2 How many of each type does this person have?
arches: _____, loops: _____, whorls: _____, composite: _____
- 3 Inspect your fingers and try to identify the patterns you have. Fill in the gaps.
arches: _____, loops: _____, whorls: _____, composite: _____.
- 4 Use a simple inepad to make a complete set of your own fingerprints on the blank form on the next page. Your teacher will show you how. Take care to clean your hands thoroughly after you finish each hand.

Elimination Fingerprint Form .. " .. " .. " .. " .. EXERCISE 3

ink pads, magnifying lens, soap

Name _____

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name _____ Taken by _____

Address _____ Rank _____

Offence _____ Date _____

Location _____ Crime Reference Number _____

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together

Dusting fingerprints .. " .. " .. " .. " .. " .. " .. " EXERCISE 4

powders, surfaces, brushes, sticky tape, black and white paper, soap

Name

Make some fingerprints on a hard, shiny black or white surface, such as a ceramic tile. How well can you see the marks? Choose a powder that is the opposite colour to the surface, and gently dust a little on the greasy mark. Blow or carefully brush off any excess powder, then use some sticky tape to lift the print. Place it on a card of a contrasting colour so that the powder shows up clearly.

- 1** Paste your best two cards in the boxes below.

Print dusted with black powder

Print dusted with white powder



- 2 a** Which powder colour do you think gave the best fingerprints? _____
b Explain why.

- 3** Do you think salt and sugar would work well as fingerprinting powders? Explain why.

- 4** Fingerprints are still the most common evidence that crime scene examiners look for at a crime scene. Suggest reasons why this is the case.

- 5** Describe the type of crime in which fingerprints would be most important.

- 6** Can you think of a crime where fingerprints would not be useful evidence? Describe the crime and say why.

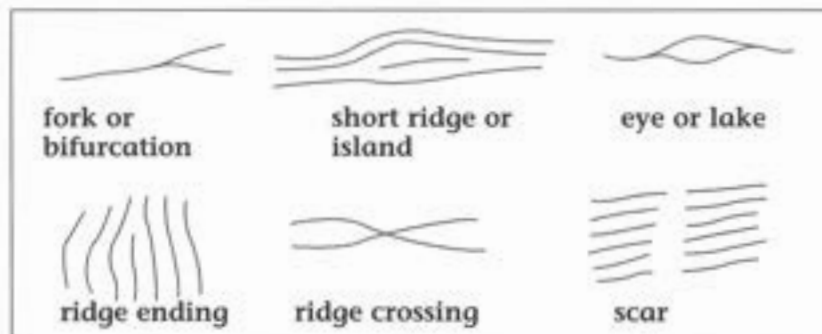
Matching fingerprints EXERCISE 5

magnifying lens

Name _____

For a fingerprint expert to declare that two fingerprints are a match (made by the same finger), not only do the patterns have to be the same, but the way the ridges join or split has to be identical.

Because of this, a fingerprint expert looks for small changes in the way the ridges flow. These changes are called minutiae or characteristics.



- 1 Label at least 10 characteristics that are visible on both the fingerprints below, then decide if the two fingerprints are a match. One characteristic has been done for you.

Rolled fingerprint

Fingerprint from crime scene



island



Do the fingerprints match? _____

- 2 There has been a theft from a cashbox at school. Mrs Kapoor's class was collecting money for a party at the end of term, but the money has disappeared. It was in Mrs Kapoor's desk and now her key is missing.

At recess time when she went to her bag, a student saw two girls from another class come out of Mrs Kapoor's room. Two boys were seen acting suspiciously near the room at lunchtime. These four students are the prime suspects.

There were some clear thumbprints on the empty cashbox, so Mrs Kapoor asked some forensic science students to find out who stole the money. Thumbprints were taken from the four suspects, Mrs Kapoor and three students (Sula, Boris and Meg) who had recently placed money in the cashbox. Your task is to discover whose thumbprints were on the cashbox and say which person you think stole the money.



- a Who left prints on the cashbox? _____
- b Who do you think stole the money? Why? _____

Making your own lip prints .. " .. " .. " .. " .. " EXERCISE 6

dark red lipstick, white paper, magnifying lens, cosmetic cleansing cream

Name _____

In this exercise, you will make your own lip prints simply by using lipstick and white paper. Then you can compare the shape and grooves of your lip prints with those of other students.

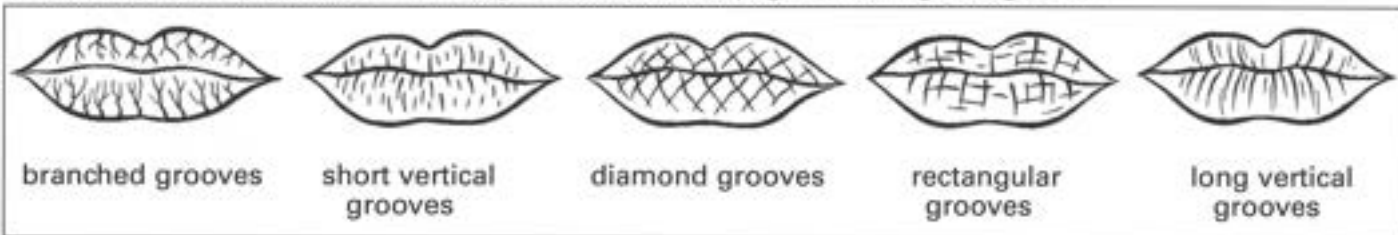
It is important that you provide your own lipstick from home. Do not share it with other students. This is to make sure that micro-organisms (germs) are not transferred from one person to another via the lipstick.

What to do

- 1 Choose a dark coloured lipstick and apply it evenly to your lips. Do not make the layer of lipstick too thick.
- 2 Take a sheet of clean white paper. Press it gently to your lips. Be careful not to slide it or press too hard, or the lip print will smudge.
- 3 Press the paper to your lips three or more times, using different places on the paper, until most of the lipstick has been transferred onto the paper.
- 4 You can reapply lipstick and repeat step 3 until you have a clear lip print that shows the grooves on your lips.
- 5 Use a cosmetic cleansing cream to remove the lipstick.
- 6 Compare your lip prints to those of others in your group or class.

My best lip print

- 1 Paste your clearest lip print in the box above right.
- 2 With freshly applied lipstick, which pressing on the paper gave you the best print? _____
- 3 Compare your lip prints with the diagrams below. (See page 15 for larger diagrams.)



To which group does your lip print belong? _____

- 4 Which type of lip print was the most common in your class? _____
- 5 Cut out the best lip prints from your group and paste them below. Label each print with the person's name and identify what type of lip print it is.

Sample lip prints

A crime is discovered PART 1

Use the information in this part to fill in Part 1 of your report.

Maddie Makura was excited as she walked home from school with Jason, her twin brother. 'I can't wait to tell Mum the results of my Maths test,' she said.

'Yeah, and I've made the football team,' Jason added.

They noticed an old, dented, blue car with its boot open, parked in front of their neighbour's house.

'That's a strange visitor for Mrs Chung to have', Jason remarked to his sister.

They continued walking towards their house. As they walked, the twins watched as a person closed the boot, walked quickly in front of the car and climbed into the driver's seat. Then a man, who had the collar on his jacket turned up, hurried out of the shrubs on the nature strip. He jumped into the passenger seat of the car, which immediately sped off.

'Those men didn't look very friendly,' Maddie observed to her brother as the car roared away from the curb, knocking over Mrs Chung's empty rubbish bin as it went.

'Men? That first one was a girl! Didn't you see her earrings?' said Jason.

'That doesn't mean it was a girl. Your friend Herman wears an earring in one ear.'

'Yeah, fair enough. But it looked like a girl to me.'

Maddie and Jason picked up the bin, then walked up their own driveway to the side door under the carport. They had a key to get in, and instructions to ring their mother at work as soon as they arrived home.

'Mum forgot to lock the door!' Jason said as he put the key back in his pocket.

'Jason, look at this!' Maddie called out from the hallway. She had rushed past her brother to let Casper in. Casper was their dog, a little Maltese terrier, and the twins had heard him barking since they turned into their street.

Jason hurried in to join Maddie, and saw the phone book lying in the middle of the hallway. The phone was off the hook and lay on top of the phone book. The drawer of the small bureau had been pulled out, and its contents strewn across the carpet.

Jason looked up at the empty space above the bureau.

'Maddie,' he exclaimed loudly, 'someone's stolen Mum's Sidney Nolan painting.'



Sidney Nolan was born in Melbourne in 1917. He studied art in Melbourne and later in Paris. During World War 2, Nolan was conscripted into the army and he served at Dimboola, a small town in Victoria, from 1942 to 1945. In 1946 he began a series of paintings about Ned Kelly the bush ranger. Later he painted other historical and legendary figures, such as Eliza Fraser and Burke and Wills. Through these paintings he became famous for the way he expressed the feeling of the country and the timelessness of the myths. He also painted a series about Central Australia. From 1950, Nolan lived mostly in the United Kingdom, but still became Australia's most internationally celebrated painter. He also designed sets for ballet and opera and provided illustrations for books. He died in London in 1992, aged 75. Sidney Nolan's paintings are now very valuable, and many of them are displayed in museums around the world.

Evidence found at 15 Sutton Street PART 2

Use the information in this part to fill in Part 2 of your report.

Once Mrs Makura, the twins' mother, and the police had arrived, the twins described what they had seen. Then Mrs Makura was interviewed by the police. They asked her who knew that she owned the Sidney Nolan painting. Mrs Makura mentioned Fiona Fitzpatrick, who cleaned the house every Tuesday. When the police asked how Fiona got into the house, Mrs Makura told them that she left a key hidden outside every Tuesday morning, and Fiona put it back in the kitchen drawer when she left. But of course today was Monday, so the spare key was still in the drawer where she always kept it. She had checked straight after the burglary, but then admitted that sometimes she didn't check whether Fiona had returned it there each Tuesday. Then she told them that Fiona was often picked up from the Makura's house by her boyfriend Daniel Davis.

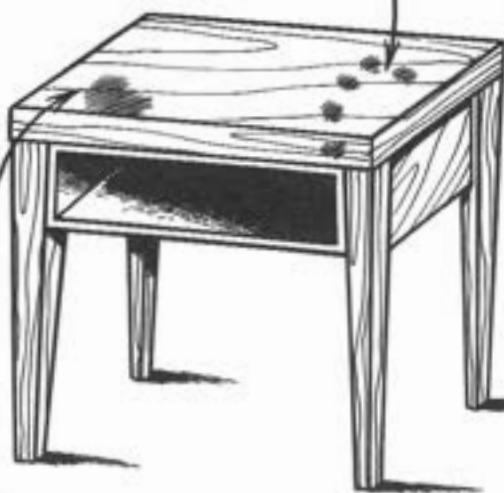
Jason told the police that when his friend Matthew Birch had stayed overnight, Matthew's older brother Brett had picked him up, and they had admired and discussed the painting before they left. Brett lived in a group house that he shared with his friend Carlos Culatto. Also living in the house was Ethan Ellis and his partner Alexa Amin, who was friends with Fiona. These six people are the first suspects the police will investigate as they try to find out who the burglars are.

The CSI team arrived shortly after the Makura family had finished speaking to the police. The crime scene examiners began searching for evidence. The first thing they established was that there was no sign of any break-in at Mrs Chung's house. It looked like the Makura home had been specifically targeted.

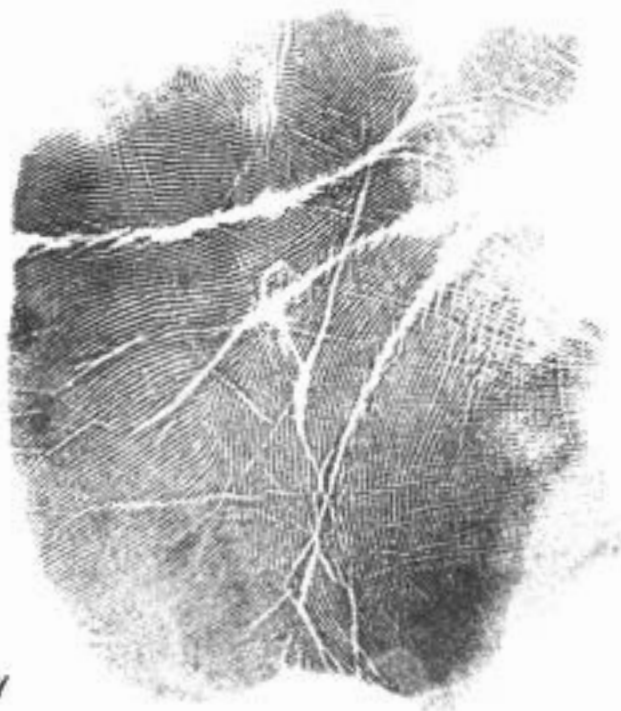
- 1 A set of fingerprints and a palm print were found on the bureau.



*Thumb print plus
4 fingerprints found
on right side of bureau*



Palm print found on left side of bureau



Evidence found at 15 Sutton Street PART 2 cont'd

- 2 A small crumpled note was found in the hallway near the bureau.

Siel Nolan original, 15 Sutton St, Take Care

- 3 Some jewellery (a bracelet) that had been in the bureau drawer was missing.



- 4 A broken dining room window was discovered. There was a shoe print and lots of glass in the garden bed outside under this window. Photographs were taken, then a plaster cast was made of the shoe print.



- 5 Some red fibres were caught on the broken glass in the window.



- 6 A lip print was found on a glass on the bench in the Makura's kitchen.



Use the information in this part to fill in Part 3 of your report.

As well as having unique fingerprints, people also have unique footprints. The only forensic drawback with footprints is that people rarely commit crimes in bare feet, so footprints are not commonly found at a crime scene. More commonly found are lip prints, which may be left on a cup or glass when someone has a drink. The analysis of lip prints is called Cheiloscopy, and it has slowly become accepted as reliable forensic evidence.

There are five basic types of lip prints.



branched grooves



diamond grooves



long vertical grooves

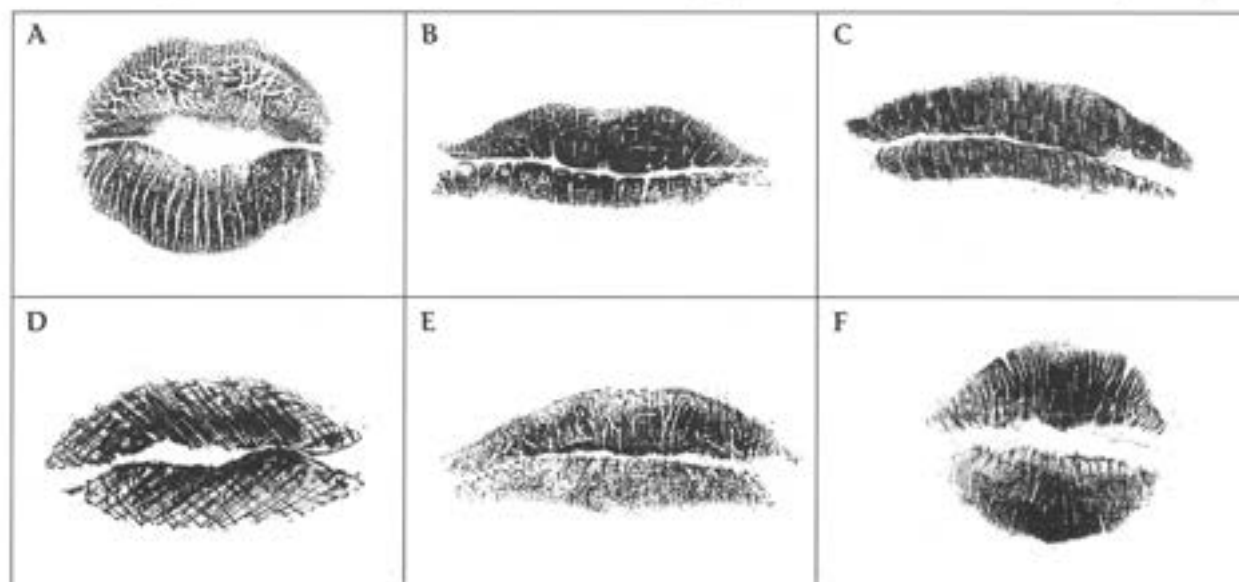


short vertical grooves



rectangular grooves

Look at the six lip prints below. For each one, write which type you think it is in the table in your report.



The lip print on the right was left on a letter. Compare it to the lip prints above. In your report, record which set of lip prints you think came from the same person.



The six lip prints above came from the suspects in the burglary case that you are investigating.

A: Alexa Amin

B: Brett Birch

C: Carlos Culatto

D: Daniel Davis

E: Ethan Ellis

F: Fiona Fitzpatrick

Compare their lip prints to those found on the glass in the kitchen (see page 14). In your report, record who you think drank from the glass in the Makura's kitchen.

Analysing paint chips and glass PART 4

Use the information in this part to fill in Part 4 of your report.

6 paint chips + 1 paint chip from crime scene, magnifying lens, tracing paper

Some scrapings of blue paint chips were left on Mrs Chung's rubbish bin and there was some broken glass on the ground nearby.

The police searched a large city car park, not far from Sutton Street. There they found six cars fitting the description given by the twins – they were all old, blue and damaged. These six cars also had broken lights.

The types of cars and owners of these cars are:

- | | |
|-----------------------------------|-----------------------------|
| A Holden Astra – Gordon Ling | D Datsun 180B – Rhea Taylor |
| B Holden Barina – Mia Ferrera | E Hyundai Excel – Lee Gough |
| C Toyota Corolla – Carlos Culatto | F Ford Laser – Gia Garner |

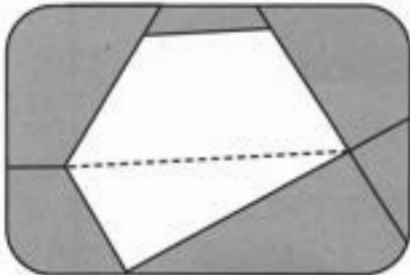
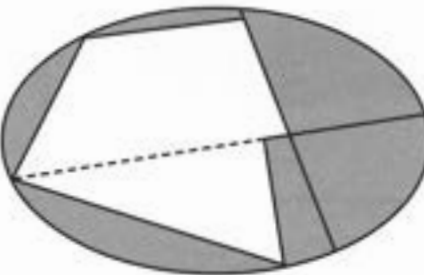
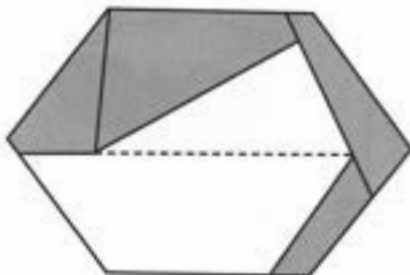
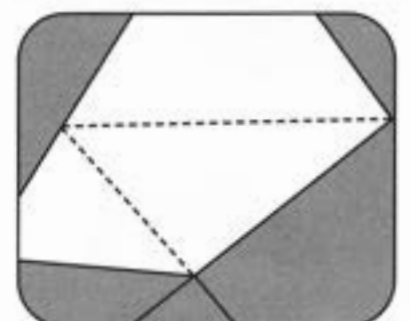
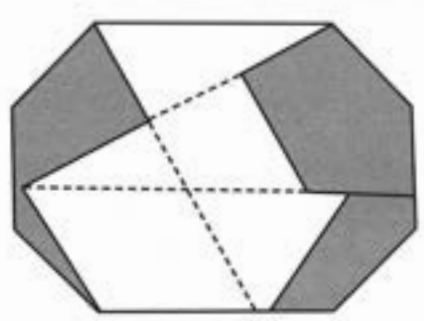
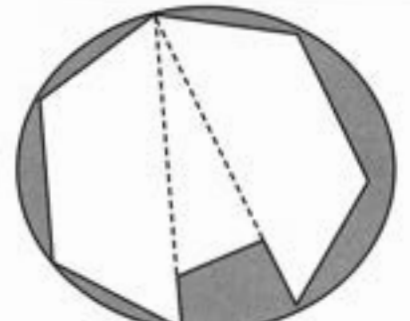
The forensic scientists took a small sample of paint from each car, then compared them to the flakes found on Mrs Chung's rubbish bin.

Use a magnifying lens to inspect the paint chips collected from the cars. Then complete the analysis table in your report, describing any layers you find in the different chips.

The police photographer took photos of the broken lights, then an artist drew matching diagrams. Use tracing paper to compare the pieces of glass found at the crime scene (beside Mrs Chung's rubbish bin) with the broken headlights below, to find out if the glass pieces came from any of these cars.

The dotted lines have been drawn in over the missing glass by the police analyst, to help you get started.



 A Holden Astra	 B Holden Barina	 C Toyota Corolla
 D Datsun 180B	 E Hyundai Excel	 F Ford Laser
Broken pieces of glass found near Mrs Chung's rubbish bin:  		

Analysing fibres PART 5

Use the information in this part to fill in Part 5 of your report.

5 red fibres + 1 red fibre from crime scene,
magnifying lens, encyclopedia or Internet

The red fibres from the crime scene need to be analysed to find out what sort of fibres they are. Once the police have discovered what sort of fibre it is, they will try to match it to a suspect.

You have been given five different fibres that are a similar colour to the fibres found at the crime scene. Inspect each sample with a magnifying lens. In your report, record which of the five samples you think matches the crime scene sample.

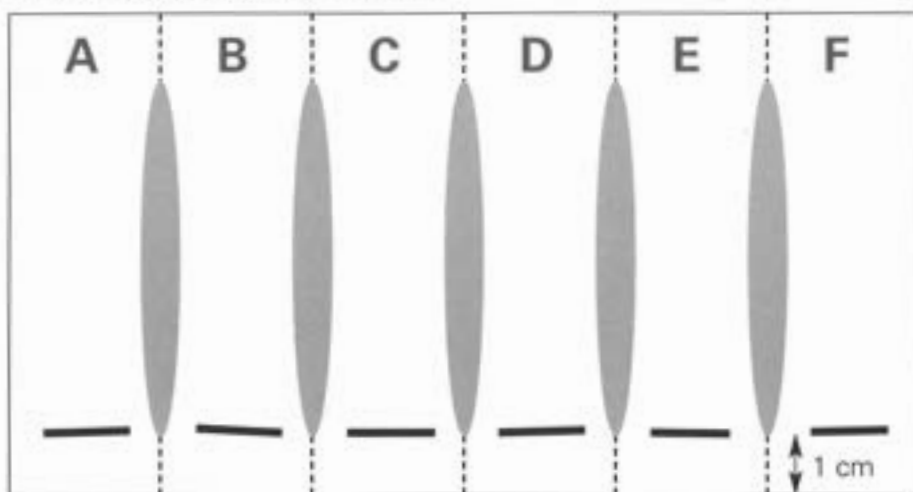
Crumpled note and chromatography PART 6

Use the information in this part to fill in Part 6 of your report.

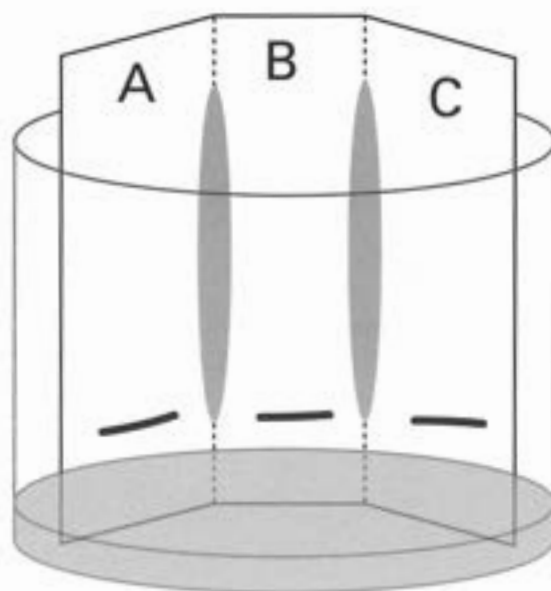
6 black pens, chromatography paper,
small beaker, stapler, crumpled note

The police visited the six suspects named by Mrs Makura and the twins. Their names are: Alexa Amin, Brett Birch, Carlos Culatto, Daniel Davis, Ethan Ellis and Fiona Fitzpatrick. Some black, felt-tipped pens were collected from their homes. Your task is to compare the ink from these pens to that on the note, using paper chromatography.

- 1 Collect a piece of absorbent paper.



- 2 Fold the paper into six as shown by the dotted lines in the diagram above.
- 3 Cut out an oval section on each fold, like those shaded in the diagram. This should leave six strips of paper attached at the top and bottom.
- 4 Draw a small line with each pen on the bottom of the paper, about 1 cm from the bottom. Label each strip at the top in pencil to show which pen was used. (The letters correspond to the initials of the six suspects.)
- 5 Stand the six strips of joined paper in a beaker that has about 0.5 cm of water in the bottom, as shown at right. Allow time for the water to soak up the paper, carrying the dyes with it.
- 6 Stand a strip of part of the crumpled note (as shown above right) in water, following the same procedure. Compare your results.



0.5 cm of water in the bottom of
the beaker, below the ink marks

Use the information in this part to fill in Part 7 of your report.

This is what the crumpled note looked like before it was cut up for the ink to be tested.

Siel Nolan original, 15 Sutton St, Take Care

Compare the handwriting on the note with the samples from the six suspects. Look at the size and style of the letters, the slope of the writing, and the spaces between letters and words. Record your observations in your report.

The quick brown fox jumps over the lazy dog. Alexa Amin	The quick brown fox jumps over the lazy dog. Brett Birch
The quick brown fox jumps over the lazy dog Carlos Culatto	The quick brown fox jumps over the lazy dog Daniel Davis
The quick brown fox jumps over the lazy dog. Ethan Ellis	The quick brown fox jumps over the lazy dog Fiona Fitzpatrick

Use the information in this part to fill in Part 8 of your report.

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.

Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name MRS SUZANNE MAKURA Taken by BROWN

Address 15 SUTTON STREET, KINGLAKE Rank D.I.

Offence BURGLARY Date 30.11.10

Location 15 SUTTON STREET, KINGLAKE Crime Reference Number 563281

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.

Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name ALEXA AMIN Taken by BROWN

Address 12 JONES AVENUE, KINGLAKE Rank D.I.

Offence BURGLARY Date 30.11.10

Location 15 SUTTON STREET, KINGLAKE Crime Reference Number 563281

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

ROLES AND REGION OF LEFT HAND				
6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.

Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name BRETT BIRCH Taken by THOMPSON

Address 12 JONES AVENUE, KINGLAKE Rank P.C.

Offence BURGLARY Date 30.11.10

Location 15 SUTTON STREET, KINGLAKE Crime Reference Number 563281

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.

Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name CARLOS CULATTO Taken by WOODS

Address 12 JONES AVENUE, KINGLAKE Rank P.C.

Offence BURGLARY Date 30.11.10

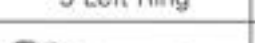
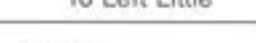
Location 15 SUTTON STREET, KINGLAKE Crime Reference Number 563281

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.

Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH
LEGITIMATE ACCESS TO A CRIME SCENE.

Name MR DANIEL DAVIS Taken by THOMPSON

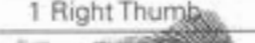
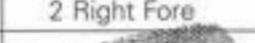
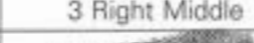
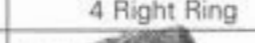
Address 12/72 FOREST ROAD, KINGLAKE Rank P.C.

Offence BURGLARY Date 30.11.10

Location 15 SUTTON STREET, KINGLAKE Crime Reference Number 563281

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.

Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name ETHAN ELLIS Taken by THOMPSON

Address 12 JONES AVENUE, KINGLAKE Rank P.C.

Offence BURGLARY Date 30.11.10

Location 15 SUTTON STREET, KINGLAKE Crime Reference Number 563281

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.

Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name FIONA FITZPATRICK Taken by WOODS

Address 8/34 SMITHERS ROAD, KINGLAKE Rank P.S.

Offence BURGLARY Date 30.11.10

Location 16 SUTTON STREET, KINGLAKE Crime Reference Number 563281

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

ROLLED IMPRESSION OF RIGHT HAND				
1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

FIONA FITZPATRICK
Right Hand



Left Hand



Use the information in this part to fill in Part 9 of your report.

The crime scene examiners took a photograph of the shoe print found in the garden before they made a plaster cast.

The police asked the six suspects to bring in their running shoes for inspection. The people in the forensics laboratory made a print of each one. These prints are shown below.



Alexa Amin



Brett Birch



Carlos Culatto



Daniel Davis

Ethan Ellis did not own any running shoes.



Fiona Fitzpatrick

Use the information in this part to fill in Part 10 of your report.

The twins had a good look at the man in the passenger seat of the car. At the police station, they looked at some books that had a number of different face shapes and other characteristics. Gradually a picture was built up of their impression of the man. They told the police that the man had dark, straight hair. It was short and receding slightly on the forehead. He had a squarish face, beard and a moustache. He wore small round glasses, had 'sticking out' ears and bushy eyebrows.

Maddie thought that the driver was a clean-shaven man with light, medium-length, wavy hair, and had flat ears and a pointed chin. He didn't wear any glasses. Jason thought the driver was a woman, as he saw that the person was wearing earrings. However, he only saw the left ear, so could only be sure of one earring.

Look at the faces below and suggest which pictures you think the police should send out for identification. Record this information in your report.



Questioning the suspects PART 11

Use the information in this part to fill in Part 11 of your report.

The six suspects were all brought into the police station for questioning.



Alexa Amin



Brett Birch



Carlos Culatto



Daniel Davis



Ethan Ellis



Fiona Fitzpatrick

The police questioned the owner of the shoe that matched the shoe prints in the garden. He said that all four men were great mates, that three of them and Alexa Amin shared a house, and that Fiona Fitzpatrick was a regular visitor to the house and also good friends with Alexa. On the day of the burglary, he had loaned his shoes to Ethan Ellis because Ethan's running shoes had worn out.

The police searched Carlos Culatto's Toyota Corolla and found a dark red woollen jumper. Carlos said that he wasn't too sure who the jumper belonged to. He had seen Ethan, Brett and even Alexa wearing it on occasions.

Carlos told police that on the day of the burglary he had loaned his car to Alexa, because her car was being repaired after a small accident a few weeks ago. Ethan had gone with Alexa in Carlos's car, because they were going to look for a new car for Alexa.

CONFIDENTIAL

RESTRICTED ACCESS

INVESTIGATIVE REPORT: THE BURGLARY ON SUTTON STREET

INVESTIGATING OFFICER(S): _____

RANK _____ OFFENCE _____

DATE _____ CRIME REFERENCE NUMBER _____

[illegible]

- 1 a** What do you think Maddie and Jason should do now, and in what order?

- b** Explain why you believe the twins should do these things.

- 2** List all the information that the twins can tell the police to help them find the culprit.

- 3** What clues or evidence do you think the crime scene examiners might look for when they come to inspect the twins' house?

- 4 List at least 5 questions that the detectives should ask the twins' mother.

INVESTIGATIVE REPORT

Evidence found at 15 Sutton Street .. " .. " .. " .. " .. REPORT PART 2

- a. Fingerprints on bureau

INVESTIGATIVE REPORT

Lip prints REPORT PART 3

- | Code | Name | Type of lip print |
|------|------|-------------------|
| A | | |
| B | | |
| C | | |
| D | | |
| E | | |
| F | | |

-

INVESTIGATIVE REPORT

Analysing paint chips and glass REPORT PART 4

- 1 Describe each paint chip sample. You could include a coloured diagram in the description.

A. Holden Astra	B. Holden Barina	C. Toyota Corolla
D. Datsun 180B	E. Hyundai Excel	F. Ford Laser
Paint chips from crime scene		

- 2 From which car do you think the paint chips found at the crime scene came?

- 3 Explain how you reached this conclusion.

- 4 What do you think the police will do now to find the culprit?

- 5 Which car do the two pieces of glass appear to come from?

- 6 Now that you know who owns the car, what can you conclude about who was the burglar? Why?

- 7 Is it possible that the owner of the car was neither the driver nor the burglar? Explain.

- 8 Once the car has been linked to the crime scene, how can that help the police to find the burglar?

- 9 What investigation or tests can the car undergo to discover who used it?

[illegible]

Type of fibre	Description of appearance
Sample 1	
Sample 2	
Sample 3	
Sample 4	
Sample 5	
Unknown type from crime scene	

- cotton _____
- wool _____
- nylon _____
- silk _____
- mohair _____
- angora _____
- polyester _____

INVESTIGATIVE REPORT

[illegible]

The police took elimination fingerprints from all six suspects and Suzanne Makura, the twins' mother. Study them carefully. For each set, identify the pattern on each of the fingers, and record these below using the following codes for the four basic patterns:

A = arch, L = loop, W = whorl, C = composite and U = unknown. Use S to record any scars.

Suspect or victim	Left hand					Right hand				
	little finger	ring finger	middle finger	fore-finger	thumb	thumb	fore-finger	middle finger	ring finger	little finger
Suzanne Makura										
	Any other comments regarding palms, fingers or size of hands									
Alexa Amin										
	Any other comments regarding palms, fingers or size of hands									
Brett Birch										
	Any other comments regarding palms, fingers or size of hands									
Carlos Culatto										
	Any other comments regarding palms, fingers or size of hands									
Daniel Davis										
	Any other comments regarding palms, fingers or size of hands									
Ethan Ellis										
	Any other comments regarding palms, fingers or size of hands									
Fiona Fitzpatrick										
	Any other comments regarding palms, fingers or size of hands									

Inspect the unidentified fingerprints and palm print found on the bureau as shown in Part 2 on page 13. Record the patterns of each finger in the table below. If the fingerprint is not present, write an X in the space. If its pattern is not clear, write "?". Compare these with the analysis of the seven sets of elimination prints that you recorded in the table above.

Prints	Left hand					Right hand				
	little finger	ring finger	middle finger	fore-finger	thumb	thumb	fore-finger	middle finger	ring finger	little finger
Fingerprints on bureau										
	Any other comments regarding palms, fingers or size of hands									
Palm print on bureau										
	Any other comments regarding palms, fingers or size of hands									

- 1 a** Whose fingerprints do you think were found on the bureau? _____
b Suggest how they came to be there. Do you think they are suspicious? Explain.

- 2 a** Whose palm print do you think was found on the bureau? _____
b Suggest how it came to be there. Do you think it is suspicious? Explain.

INVESTIGATIVE REPORT

Shoe prints REPORT PART 9

- 1 Whose shoe matches the impression found in the garden bed?
- 2 Does this mean that this person committed the crime? Explain your answer.
- 3 What additional information would be useful before the police could charge the owner of this shoe with the crime?

[illegible]

- 1** Which identikit picture(s) should the police send out to the media for identification? Why?

Questioning the suspects .. " .. " .. " .. " .. " .. " .. " .. " .. " .. " .. " REPORT PART 11

- 1** Which two suspects match the twins' descriptions most closely?
 - a** Passenger of the old blue car _____
 - b** Driver of the old blue car _____
- 2** Who do you think left the shoe prints in the garden? _____
- 3** Who do you think may have worn the dark red jumper found in Carlos's car on the day of the burglary? _____

INVESTIGATIVE REPORT

Summary of all the evidence REPORT PART 12

- 1 The fingerprints found on the bureau belonged to _____ and were left there during _____
- 2 The palm prints on the bureau were left by _____. This was most likely when he was _____
- 3 The probable owner of the shoes that left their impressions in the garden below the broken window was _____. These shoes were most likely worn by _____ during the burglary.
- 4 The broken glass from the window had mostly fallen on the outside of the house. Can you explain why this might be the case? Suggest who might have broken the window and why.

- 5 The crumpled note was probably written by _____ using a pen owned by _____. The purpose of the note was to _____

- 6 The red fibres probably came from _____

- 7 The blue paint chips found on Mrs Chung's rubbish bin are believed to have come from the _____ car belonging to _____ having been driven by _____.
- 8 The lip prints on the glass in the kitchen were most likely left by _____. This probably occurred when _____

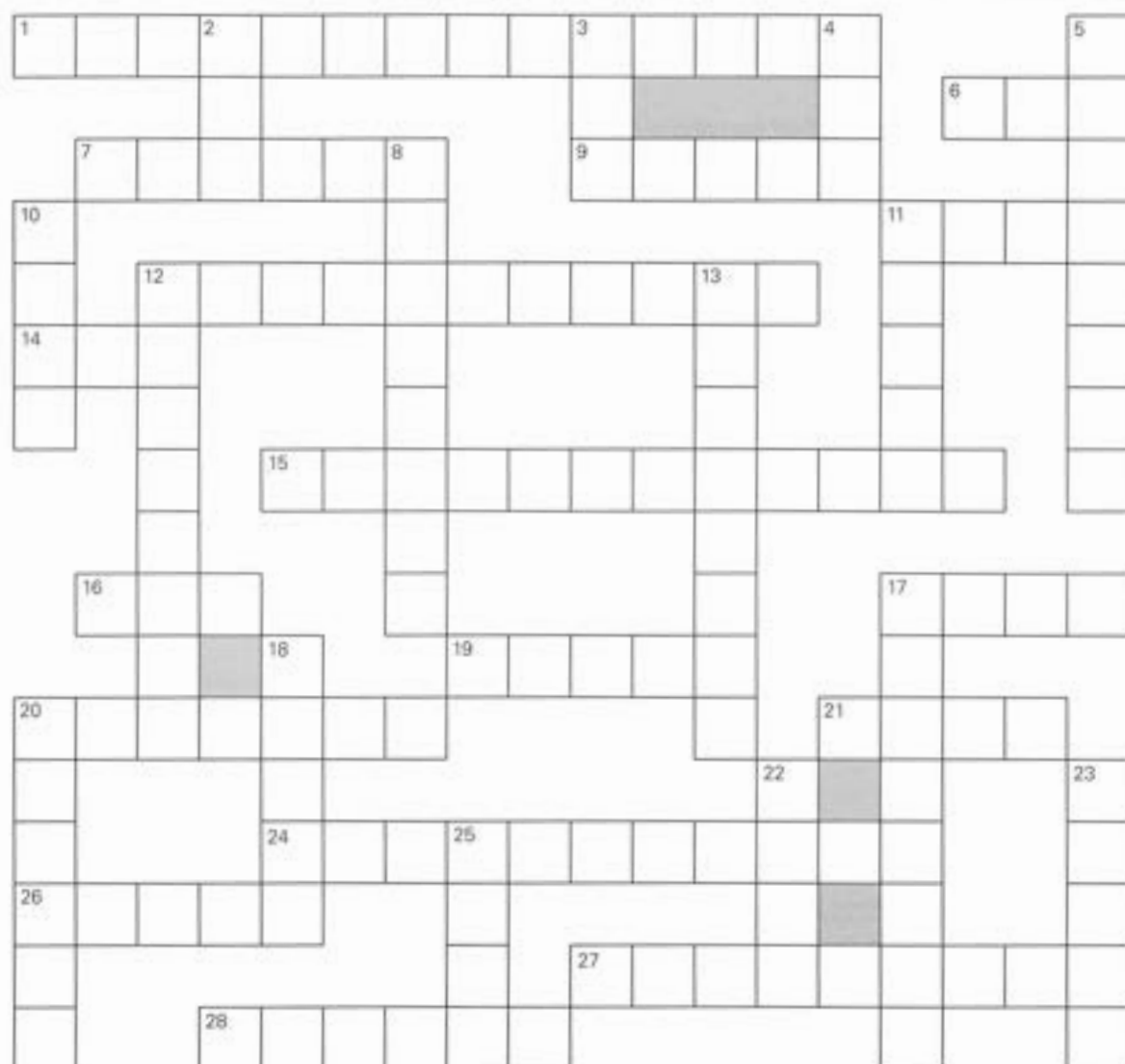
INVESTIGATIVE REPORT

9 Describe exactly how you think the crime was carried out. Include who did what, when and why.

Try to explain the following:

- a** Why the twins did not see the thief carrying out their mother's painting when he climbed in the car.
b How involved you think the driver of the car was.
c Why was the glass from the broken window on the outside of the house?

Forensic Science crossword



Across

1. A technique used to compare dyes in pen inks
6. Means of transport used by burglar to get away from the crime scene
7. People employed to help citizens
9. The surname of the suspect whose first name is Daniel
11. Another body part that has furrows and ridges similar to those on the fingers
12. The type of fingerprints taken by police
14. The number of suspects the police investigated for this crime
15. Evidence left behind when a person touches things
16. The writing fluid found in pens
17. The number of basic fingerprint patterns
19. A fingerprint pattern where the ridges consist of many ovals of different sizes
20. Another name for a thief
21. The simplest fingerprint pattern
24. A very famous Australian artist
26. The suspect Ethan's surname
27. The name given to the drawing of a face put together from a witness's description
28. Useful crime evidence that comes from the clothes that a person is wearing

Down

2. Substance produced by the skin and left behind in fingerprints
3. The colour of the fibres found caught on the broken window glass
4. If you see a crime being committed, should you call the police?
5. Someone who commits a crime
8. Another name given to 'clues' found at a crime scene
10. This is made by police if a shoe print is found
11. This can be left behind if a car collides with a stationary object like a tree or pole
12. The person who searches a crime scene is called a 'Crime Scene ____'
13. An item of clothing worn by forensic people when they are working within a crime scene
17. The type of science that is used to solve crimes
18. Evidence left behind when a window is broken
20. A type of desk
22. Before the burglary at 15 Sutton Street was carried out, the criminals hatched a ____
23. The first name of the suspect whose surname was Birch
25. This was used to tell the burglar where to go and what to steal from the Makura house

Exercise 1 – Searching the crime scene

- 1 To ensure they do not contaminate the crime scene with their own fingerprints, hairs or fibres.
- 2 To ensure that the position where evidence was found is recorded.
- 3 A torch helps the CSE to see more clearly.
- 4 Shoe prints or tyre prints are photographed before a cast is made.
- 5 Small items are collected with tweezers and placed in sealed, labelled bags, then sent to the laboratory for further testing.

Exercise 2 – Fingerprints

- 2 four arches, three loops, two whorls, one composite

Exercise 4 – Dusting fingerprints

- 2 Probably black is clearer.
- 3 Sugar and salt would not work well because the crystals are too large to stick to the oil from the fingerprint.
- 4 Fingerprints are very common because many people pick up objects with their hands without realising that they are leaving a small amount of oil and sweat behind.
- 5 Fingerprints would be more important in crimes where a person has handled objects or parts of the crime scene with their hands.
- 6 Fingerprints are less useful at crimes where there is no physical evidence, such as computer crimes, arson, or where gloves have been worn.

Exercise 5 – Matching fingerprints

- 1 The two fingerprints match.
- 2 a Mrs Kapoor, Boris and Meg and suspect 4 left thumbprints on the cashbox.
b Suspect 4 must have stolen the money.

Part 1 – A crime is discovered

- 1 a The twins should move outside immediately and go to a neighbour's house where they should ring the police and then their mother.
b The twins should leave the house because the burglar might be dangerous and may still be inside the house. The police should be called because they are trained to catch people like this. Their mother needs to know that they are safe and what has happened.
- 2 The twins can tell the police when they discovered the theft, give a description of the strangers and the car, and describe other details they saw such as the car hitting the bin and what they saw when they walked inside their house.
- 3 CSEs will look for how the thief entered, fingerprints, hairs and fibres, shoe prints and anything out of place.
- 4 Questions could include – When did you leave home today? Who was last to leave? Was the door locked? Who knows that you own the painting? Who has a key to your house? Have you seen anyone strange or anyone acting suspiciously lately?
- 5 Mrs Chung and any other neighbours. Ask them if they have seen anything strange recently.
- 6 A type of desk.
- 7 Answers could include: Sidney Nolan became famous after 1946; he is described as Australia's most internationally celebrated painter; all of his paintings are now very valuable; his paintings are displayed in museums around the world.

Part 2 – Evidence found at 15 Sutton Street

- 1 a Dust, photograph, lift and send to lab for matching.
b Photograph, collect glass and send to lab for possible matching in the future.
c Photograph, make a cast to compare with shoe prints from suspects in the future.
d Collect, send to lab for handwriting comparison and ink testing.
e Collect and send to lab for matching in the future.
f Photograph glass, dust and lift for later matching with suspect's lip prints.
- 2 If any of above evidence matches that from a suspect, it links that person to the crime scene, so the police have someone to question about their movements at the time of the theft.

Part 3 – Lip prints

- 1 A – Alexa Amin – branched and long vertical; B – Brett Birch – rectangular; C – Carlos Culatto – short vertical grooves; D – Daniel Davis – diamond; E – Ethan Ellis – branching; F – Fiona Fitzpatrick – long vertical
- 2 The lip print on the letter matches Fiona Fitzpatrick.
- 3 The lip print found on the glass belonged to Alexa Amin.

Part 4 – Analysing paint chips and glass

- 2 The paint chips from the crime scene match C, the Toyota Corolla.
- 3 The coloured layers on both chips are the same and in the same order.
- 4 The police will check who owns this car and who was driving it on the day the crime was committed.
- 5 Carlos Culatto's Toyota Corolla.
- 6 Nothing can be concluded about the burglar, but the car belonging to Carlos Culatto was most likely the one that hit Mrs Chung's rubbish bin.
- 7 Yes, this is quite possible. He needs to be questioned about who was driving his car that afternoon.
- 8 This evidence is important because it leads to the questioning of Carlos Culatto, who should know who was driving his car at the time.
- 9 CSEs should look for fingerprints in the car.

Part 5 – Analysing fibres

- 1 Wool
- 2 They will keep the fibre to match with similar fibres collected from the suspects.
- 3 Cotton is natural, from the cotton plant; wool is natural, from sheep; nylon is synthetic; silk is natural, from mulberry silk worms; mohair and angora are natural, wool from goats; polyester is synthetic.

Part 6 – Crumpled note and chromatography

- 1 The ink on the crumpled note most closely matches pen D.
- 2 No, this does not prove that the owner of pen D actually wrote the note.
- 3 That the crumpled note was most likely written using pen D or a pen of the same type.
- 4 Handwriting comparison would be most helpful here.

Part 7 – Handwriting

- 1 Fiona Fitzpatrick.
- 2 Same style in writing letters r, s, t; similar spacings and slope.
- 3 Fiona may have written the note to give directions to the burglars to find the painting.
- 4 Fiona may suggest her handwriting was forged by another person to implicate her, or that the similarity in handwriting is just a coincidence.

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing the highest quality learning outcomes with instant usability.

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Forensic Science Hypothetical: The Burglary in Sutton Street

This third title in the **Forensic Science Hypothetical** series by best-selling author Jeanette Jolley provides photocopiable worksheets containing all the information and evidence that students need to solve a burglary. As they work through the scenario, students record their findings in a case file, developing logical conclusions based on evidence. This is an ideal resource for exploring the practical application of science knowledge and skills. Students will enjoy the opportunity to detect who committed the crime as they use key elements of forensic science such as collecting evidence, examining witness statements and testing materials using chromatography.

No previous study or knowledge of forensic science is necessary, as background information and introductory tasks are provided. The activities require minimal teacher preparation and answers have been included for time efficient marking.

Topics include

Discovering a crime
Searching a crime scene
Collecting, analysing and recording evidence
Collecting and matching fingerprints
Comparing shoe prints and lip prints
Investigating evidence – glass, fibres and paint chips

Handwriting and forgeries
Identikit sketches
Testing inks with chromatography
Analysing suspect and witness statements
Questioning suspects and developing logical conclusions based on evidence
Exploring motives

Associated titles in the Instant Lessons series

- Forensic Science Hypothetical
 - The Kidnapping of Jamie Lawson
 - The Death of Barnaby Hill
- Forensic Science for High Schools (Books 1, 2, 3, 4)
- Literacy in Science (Books 1, 2, 3, 4)
- Legends of Science (Books 1, 2)

